

Venkata Saratchandra Patnaik Markonda Patnaikuni

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EDUCATION

Master of Science in Computer Science (CGPA: 3.78/4)

December 2025

Arizona State University, Tempe, AZ.

Bachelor of Technology in Computer Science (CGPA: 9.11/10)

May 2022

Christ University - Faculty of Engineering, India.

TECHNICAL SKILLS

Programming & Automation: Python, Java, SQL, Kotlin, JavaScript, C

Computer Vision: OpenCV, NumPy, Matplotlib, Image resizing, histogram equalization, morphological operations, edge detection (Sobel/Prewitt), line detection (Hough Transform), distance transform, thresholding (Otsu)

Machine Learning: Supervised learning, feature extraction, handling imbalanced datasets, classification

Databases: MySQL, PostgreSQL, Firebase, MongoDB

Development, Cloud & Tools: AWS (S3, EC2, Lambda), Docker, Jenkins, Agile, GitHub, REST APIs, Automation Scripting

PROFESSIONAL EXPERIENCE

Software Engineer: Amagi Media Labs, Bengaluru, India (Media)

August 2022 - November 2023

- Designed onboarding solutions for content creators, optimized **video ingestion** and distribution, and automated customer onboarding using Terraform, reducing manual setup time by 90%.
- Developed Python scripts for **video preprocessing**, reducing processing time by 37%, automated **API-based video access** verification, and ensured global content delivery via CDNs, reducing failures by 95%.
- Managed 15+ containerized microservices on AWS EKS with Kubernetes and Docker, automating deployments with Linux scripting, and improving weekly deployment frequency by 30%.

RESEARCH EXPERIENCE

Computer Vision Laboratory - CSHOI, Christ University, Bengaluru, India

August 2021 - December 2021

- Developed and implemented **11 computer vision experiments** using **Python and OpenCV**, mastering image processing techniques like **resizing, histogram equalization, and morphological operations** for applications in computer vision.
- Feature extraction** through **edge detection (Sobel/Prewitt), line detection (Hough Transform), and distance transform**, optimizing algorithms for accuracy in structured environments, relevant to multimodal learning.
- Applied segmentation techniques**, including **Otsu's thresholding**, and conducted **quantitative analysis** with **Matplotlib**, building a strong foundation for **vision-language integration** and human-robot interaction research.

PROJECTS

Handling Imbalanced Data on Bank Churn Prediction (Machine Learning & Data Analytics)

- Developed a **machine learning model** to **predict** customer churn using **Python**, addressing imbalanced datasets with **SMOTE** and **undersampling**, improving model accuracy by 15%.
- Performed **data cleaning** and **visualization** with **Pandas** and **Matplotlib**, extracting key features for classification that were relevant to multimodal data processing in vision research.

Automated Online Attendance Bot (Automation Anywhere)

- Developed an **RPA bot** using **Automation Anywhere a2019** to automate Webex attendance tracking, processing participant data with a 95% reduction in manual effort, and reclaiming 10 hours weekly.
- Implemented a 50-minute threshold for accurate data extraction, enhancing efficiency in data handling, applicable to real-time vision and multimodal data processing.

RESEARCH INTERESTS

- Vision-language integration for human-robot interaction
- Multimodal learning for video understanding in robotics
- Scalable computer vision pipelines using cloud infrastructure